

Implementation of Cross-Platform Data Transmission Systems: A Comparative Study of Wired Serial and Bluetooth Low Energy (BLE) Bridging using nRF5340

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Abstract—This paper presents the design and implementation of a cross-platform bidirectional data transfer system, enabling seamless telemetry and file exchange between a Linux host and an Android mobile environment. The study contrasts a baseline wired serial connection against a wireless embedded bridge utilizing the Nordic Semiconductor nRF5340 SoC.

In the wireless implementation, the nRF5340 functions as a protocol converter, bridging USB-CDC (ACM) serial data to Bluetooth Low Energy (BLE) packets via the Nordic UART Service (NUS). To mitigate data corruption caused by control character interpretation in the UART stack, a Base64 encoding scheme was integrated, successfully enabling the transmission of binary assets (images, PDFs) with 100% data integrity.

I. HARDWARE REQUIREMENTS

- Development Board: Nordic Semiconductor nRF5340 DK.
- Smartphone: Android device with nRF Connect for Mobile and Serial Bluetooth Terminal App.
- PC: Linux machine

II. SOFTWARE AND TOOLCHAINS INSTALLATIONS

A. 1. Update & Install System Dependencies

First, update the package manager and install the core dependencies required by the Zephyr build system (CMake, Ninja, Python, etc.).

```
sudo apt update
sudo apt upgrade -y
sudo apt install --no-install-recommends -y \
    git cmake ninja-build gperf \
    ccache dfu-util device-tree-compiler \
    wget \
    python3-dev python3-pip \
    python3-setuptools python3-tk \
    python3-wheel xz-utils file \
    make gcc gcc-multilib g++-multilib \
    libstdl2-dev
```

B. 2. Set up Python Virtual Environment & Install West

Modern Linux distributions enforce PEP 668, which restricts global 'pip' installations. The Zephyr build system requires a dedicated Python virtual environment.

```
# Create a virtual environment in the home
# directory
python3 -m venv ~/zephyr-venv

# Activate the virtual environment
source ~/zephyr-venv/bin/activate

# Install west inside the virtual
# environment
pip3 install -U west
```

Note: You must run source ~/zephyr-venv/bin/activate every time you open a new terminal to build the project.

C. 3. Initialize nRF Connect SDK (v3.2.2)

Create a workspace directory and initialize the specific SDK version used for this project.

```
mkdir ~/ncs
cd ~/ncs

# Initialize the SDK pointing to version
# v3.2.2
west init -m
https://github.com/nrfconnect/sdk-nrf
--mr v3.2.2

# Download the code
west update

# Export CMake package for Zephyr
west zephyr-export
```

D. 4. Install Python Dependencies

With the virtual environment activated, install the specific Python requirements for the Zephyr kernel, the nRF SDK, and the bootloader.

```

pip3 install -r
    zephyr/scripts/requirements.txt
pip3 install -r nrf/scripts/requirements.txt
pip3 install -r
    bootloader/mcuboot/scripts/requirements.txt

```

E. 5. Install the Zephyr SDK Toolchain

Download and install the cross-compiler toolchain (Zephyr SDK 0.16.8) required to compile code for the nRF5340 on NCS v3.2.x.

```

# Download the Zephyr SDK Bundle
cd ~
wget
    https://github.com/zephyrproject-rtos/sdk-ng/releases/download/v0.16.8/zephyr-sdk-0.16.8_linux-x86_64.tar.xz

# Extract the archive
tar xvf
    zephyr-sdk-0.16.8_linux-x86_64.tar.xz

# Run the setup script
cd zephyr-sdk-0.16.8
./setup.sh -c

```

F. 6. Install nRF Command Line Tools

Finally, install the J-Link drivers and nrfjprog to enable flashing the board over USB.

```

# Navigate to the download directory
cd ~/Downloads

# Install the .deb package (replace
    filename with actual version)
sudo dpkg -i nrf-command-line-tools*.deb

# Fix any missing dependencies
sudo apt install -f

```

G. Verification

Verify the installation by checking the versions of the flashing and build tools.

```

nrfjprog --version
west --version

```

III. PROJECT EXECUTION PROCEDURE

This section details the procedure to compile, flash, and test the `peripheral_uart` sample application using the command line interface.

A. 1. Workspace Initialization

Before building, ensure the workspace is correctly set up. If you have not created a specific project folder yet, navigate to your NCS root directory.

```

cd ~/ncs
# Ensure environment variables are set
    (optional but recommended)
source zephyr/zephyr-env.sh

```

B. 2. Building the Firmware

Use the `west build` command to compile the sample code. We target the `peripheral_uart` sample located in the Bluetooth folder.

Target Board: `nrf5340dk/nrf5340/cpuapp` (Application Core).

```

# Navigate to the main folder
cd ~/ncs

# Build the peripheral_uart sample
west build -b nrf5340dk/nrf5340/cpuapp
    nrf/samples/bluetooth/peripheral_uart

```

Note: If the build fails due to cached configurations, append the `--pristine` flag `-p` to clean the build directory.

```

west build -p -b nrf5340dk/nrf5340/cpuapp
    nrf/samples/bluetooth/peripheral_uart

```

C. 3. Flashing the Board

Once the build completes successfully, flash the binary to the nRF5340 development kit.

- 1) Connect the nRF5340 DK to the PC via the **IMCU USB** port (Interface MCU).
- 2) Run the flash command:

```

west flash

```

If successful, the terminal will display `Flashing completed`.

D. 4. Testing Data Transfer

To verify functionality, we establish a connection between the PC terminal and a smartphone.

1) **A. PC Setup (Receiver):** Open a serial terminal to view data sent from the phone.

- **Command:** Use `screen` or `minicom` to listen to the board's serial output.
- **Baud Rate:** 115200.

```

# List available USB devices to find the
    port (e.g., /dev/ttyACM0)
ls /dev/ttyACM*

# Open the serial connection
sudo screen /dev/ttyACM0 115200

```

2) **B. Smartphone Setup (Sender):**

- 1) Open the **nRF Connect for Mobile** app on Android.
- 2) Scan for devices and locate **Nordic_UART_Service**.
- 3) Tap **Connect**.
- 4) Navigate to the **UART** tab.
- 5) Type a text message (e.g., "Hello Linux") and send it.

3) **C. Result:** The text typed on the smartphone should immediately appear in the Linux terminal window. Conversely, typing in the Linux terminal will send text to the smartphone app logs.

IV. FILE TRANSFER : LINUX TERMINAL & ANDROID SERIAL APP

This section details the procedure to verify bidirectional text transfer using a Linux terminal on the PC side and the **Serial Bluetooth Terminal** app on an Android device.

A. 1. Linux Setup (Receiver/Sender)

On the Ubuntu machine, we interact with the nRF5340 board through its virtual serial port.

- 1) Connect the nRF5340 DK to the PC via the **IMCU USB** port.
- 2) Open a terminal window and list the USB devices to identify the board (usually `tttACM0` or `tttACM1`).

```
ls /dev/ttyACM*
```

- 3) Open the serial connection using `screen` or `minicom` with a baud rate of **115200**.

```
# Replace tttACM0 with your actual port
sudo screen /dev/ttyACM0 115200
```

- 4) The terminal screen will clear and wait for input. (Press **Reset** on the board if you want to see the boot log).

B. 2. Android Setup (Serial Bluetooth Terminal)

We use a dedicated terminal app to act as the Bluetooth Central device.

- 1) **Install App:** Download and install **Serial Bluetooth Terminal** (by Kai Morich) from the Google Play Store.
- 2) **Open App:** Launch the application.
- 3) **Scan:**
 - Tap the menu (three lines) in the top-left corner.
 - Select **Devices**.
 - Tap the **Bluetooth LE** tab (important: do not use the Classic Bluetooth tab).
 - Tap **Scan** at the top.
- 4) **Connect:** Locate the device named **Nordic_UART_Service** in the list and tap it to connect. The terminal screen should show "Connected".

C. 3. Verifying File Transfer (Text Streaming)

This section details the procedure to transfer entire text files between the Android device and the Linux machine using the Nordic UART Service.

1) *Smartphone to Linux (Sending a File):* To send a text file stored on the phone to the Linux terminal:

- 1) Ensure the devices are connected via the **Serial Bluetooth Terminal** app.
- 2) Tap the **Menu** icon (three lines) in the top-left corner or the **Upload/Folder** icon (if available on the toolbar).
- 3) Select **Send File**.
- 4) Browse and select a `.txt` file from your phone's storage.
- 5) **Result:** The content of the file will stream immediately into the open screen session on your Linux terminal. This can be saved as a file in the PC.

2) *Linux to Smartphone (Sending a File):* To send a text file from the Linux machine to the phone, we cannot just type in the terminal. We must pipe the file directly to the device port.

- 1) Open a **new** terminal window (keep the screen session open in the first window to monitor the connection).
- 2) Create a dummy text file to send:

```
echo "This is a test file sent from
Linux." > test_file.txt
```

- 3) Use the `cat` command to redirect the file into the USB serial port:

```
# Replace tttACM0 with your actual port
sudo sh -c 'cat test_file.txt >
/dev/ttyACM0'
```

- 4) **Result:** The content of `test_file.txt` will appear in the **Serial Bluetooth Terminal** app on your phone (usually in blue).

V. TRANSFERRING BINARY FILES (IMAGES/PDFs) VIA BASE64

The Nordic UART Service (NUS) is designed for character-based data streams. Attempting to send raw binary files (e.g., `.jpg`, `.png`, `.pdf`) directly will result in data corruption due to control characters and null bytes. To transfer these files, we must encode them into a text-safe format using Base64.

A. 1. Concept Overview

The process involves three stages:

- 1) **Sender (Linux):** Encodes the binary file into a Base64 text string.
- 2) **Transmission:** The text string is sent over the USB-to-Bluetooth bridge.
- 3) **Receiver (Smartphone/PC):** Receives the text string and decodes it back into the original binary format using standard Base64 decoding tools.

B. 2. Technical Constraints

- **Overhead:** Base64 encoding increases the file size by approximately 33%.
- **Speed:** Bluetooth Low Energy (BLE) has a limited bandwidth (approx. 10-20 kB/s for NUS). Transferring large files (e.g., > 1MB) may take several minutes.
- **Buffer:** Ensure the receiving device does not sleep during transfer, as BLE connections may drop if the radio is idle for too long.

VI. FILE TRANSFER BETWEEN 2 DEVICES VIA BLUETOOTH WITHOUT USING NRF5340 BLE

Switch on bluetooth on both devices.

A. Installing the Packages

```
sudo apt update
sudo apt install bluez bluetooth
bluetooth-obex obexftp gnome-bluetooth
sudo systemctl start bluetooth
```

B. Connecting the sender to the receiver

On the sender device:

```
bluetoothctl scan on
```

Note the MAC address: AA:BB:CC:DD:EE:FF

```
bluetoothctl
pair AA:BB:CC:DD:EE:FF
trust AA:BB:CC:DD:EE:FF
connect AA:BB:CC:DD:EE:FF
exit
```

```
bluetooth-sendto --device=AA:BB:CC:DD:EE:FF
/path/to/file1.txt /path/to/file2.jpg
/path/to/file3.png
```

To automate this process:

Save the following as bt-multi-send.sh

```
#!/bin/bash
DEVICE="AA:BB:CC:DD:EE:FF" # Replace with
MAC Address
DIR_PATH="$1" # Pass dir as arg, e.g.,
./bt-multi-send.sh /path/to/files

if [ -z "$DIR_PATH" ]; then
    echo "Usage: $0 <directory>"
    exit 1
fi

# Ensure paired
bluetoothctl connect $DEVICE > /dev/null
2>&1

for file in "$DIR_PATH"/*; do
    if [ -f "$file" ]; then
        echo "Sending $file..."
        bluetooth-sendto --device=$DEVICE
"$file"
    fi
done
```

Make it executable:

```
chmod +x bt-multi-send.sh
```

Run this command:

```
./bt-multi-send.sh /path/to/files
```

VII. INDUSTRY APPLICATIONS OF NORDIC SoC'S

Nordic Semiconductor SoCs are widely adopted in commercial IoT sectors. Notable examples include:

- **Wearables:** The *Ultrahuman Ring Air* and *Dayton Smart-watch* utilize the nRF52 series for biometric tracking (ECG, Sleep Index).
- **Medical Healthcare:** Devices such as *PulseTape* utilize Nordic chips for continuous ECG monitoring, prioritizing low power consumption for extended battery life.
- **Smart Home:** Significantly, the *eCozy 2.0 Smart Thermostat* employs the **nRF5340**—the exact SoC utilized in this project—to handle complex connectivity protocols like Matter and Bluetooth Mesh.

VIII. CONCLUSION

This project successfully demonstrated the implementation of a robust, bidirectional wireless data transfer system utilizing the Nordic Semiconductor nRF5340 System-on-Chip. By configuring the nRF5340 as a bridge between a USB-CDC (Serial) interface and the Bluetooth Low Energy (BLE) Nordic UART Service (NUS), we established a seamless communication link between a Linux-based PC and an Android mobile environment and between 2 Linux-based PC.

- **Firmware Implementation:** The firmware was successfully built and flashed using the Zephyr RTOS command-line toolchain (west), demonstrating a professional, reproducible development workflow without reliance on heavy GUI IDEs.
- **Data Integrity:** While the UART protocol is inherently text-based, we successfully implemented a Base64 encoding strategy. This enabled the error-free transmission of complex binary files (images, PDFs) over the link, proving the system's versatility beyond simple telemetry.

A. Final Verdict

The comparative analysis between direct wired connections and the nRF5340-based solution highlights the flexibility of BLE for embedded applications. While wired connections offer higher raw throughput, the nRF5340 implementation provides a reliable, low-power wireless alternative suitable for IoT remote debugging, field telemetry, and file exchange in environments where physical cabling is impractical. The system stands as a functional proof-of-concept for modern embedded wireless bridging.

REFERENCES

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